

Uji Akurasi

Sample 1

Classification Metrics

...

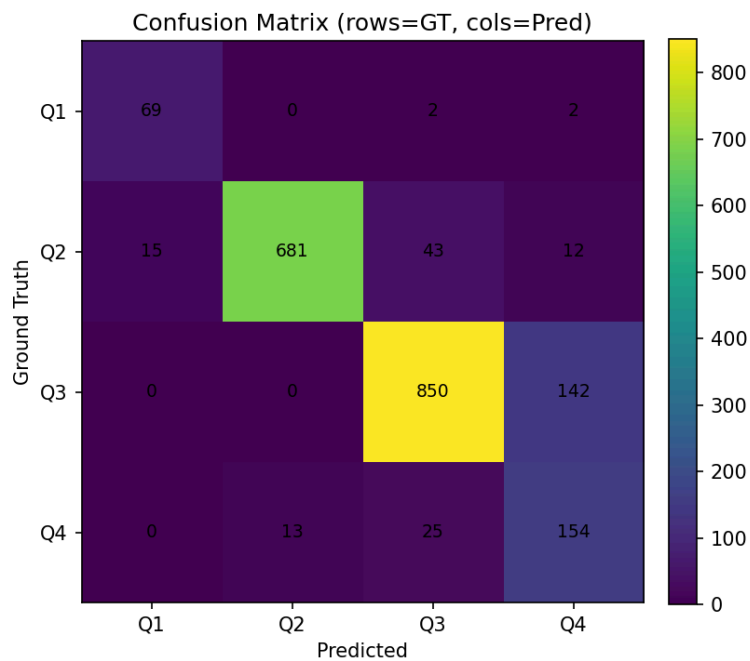
```
{
  "per_class": {
    "Q1": {
      "precision": 0.8214285714285714,
      "recall": 0.9452054794520548,
      "f1": 0.8789808917197452,
      "support": 73
    },
    "Q2": {
      "precision": 0.9812680115273775,
      "recall": 0.9067909454061251,
      "f1": 0.942560553633218,
      "support": 751
    },
    "Q3": {
      "precision": 0.9239130434782609,
      "recall": 0.8568548387096774,
      "f1": 0.8891213389121339,
      "support": 992
    },
    "Q4": {
      "precision": 0.4967741935483871,
      "recall": 0.8020833333333334,
      "f1": 0.6135458167330677,
      "support": 192
    }
  },
  "averages": {
    "micro": {
      "precision": 0.8735059760956175,
      "recall": 0.8735059760956175,
      "f1": 0.8735059760956175
    },
    "macro": {
      "precision": 0.8058459549956491,
      "recall": 0.8777336492252977,
      "f1": 0.8310521502495412
    },
    "weighted": {
```

```

    "precision": 0.9007962881788203,
    "recall": 0.8735059760956175,
    "f1": 0.8823893156811126
  }
},
"accuracy": 0.8735059760956175,
"support_total": 2008
}
...

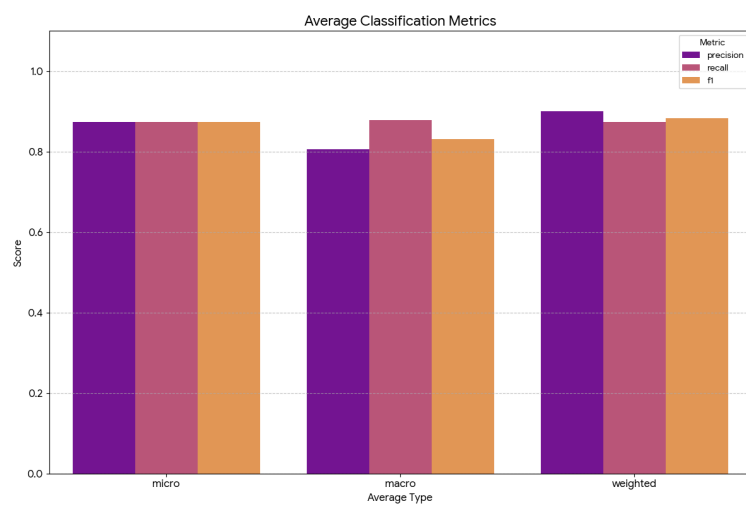
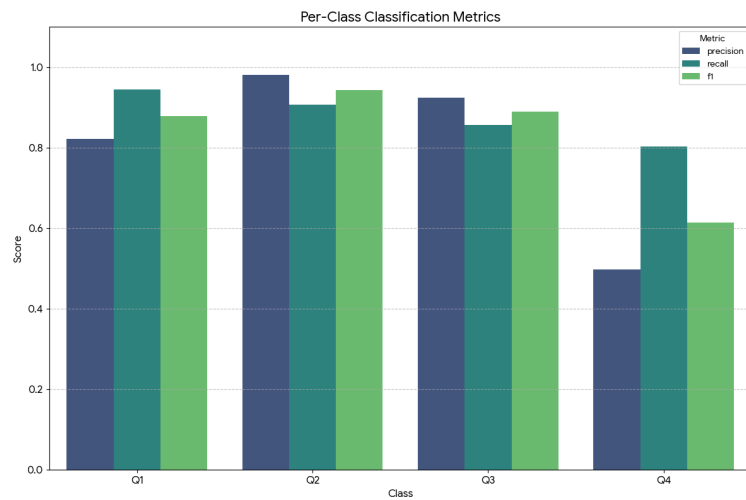
```

Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	69	0	2	2
Q2	15	681	43	12
Q3	0	0	850	142
Q4	0	13	25	154

Precision, Recall and F1



Sample 2

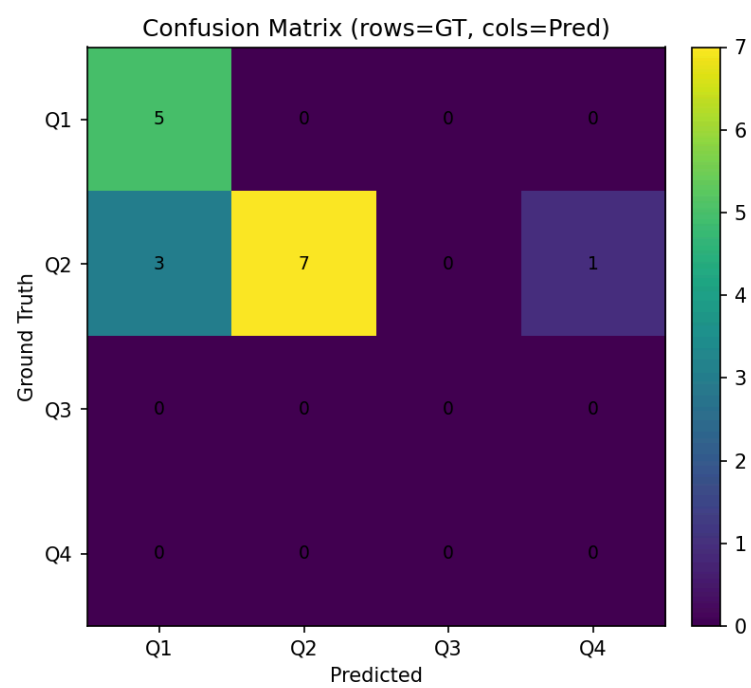
Classification Metrics

...

```
{
  "per_class": {
    "Q1": {
      "precision": 0.625,
      "recall": 1.0,
      "f1": 0.7692307692307693,
      "support": 5
    },
    "Q2": {
      "precision": 1.0,
      "recall": 0.6363636363636364,
      "f1": 0.7777777777777778,
      "support": 11
    },
    "Q3": {
```

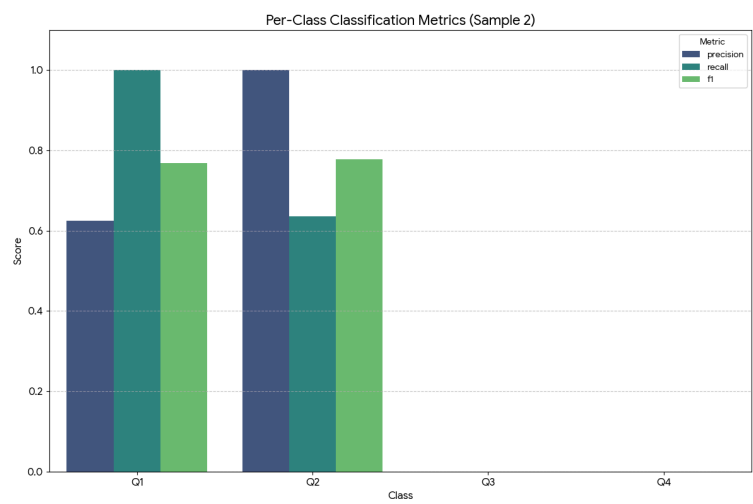
```
    "precision": 0.0,  
    "recall": 0.0,  
    "f1": 0.0,  
    "support": 0  
  },  
  "Q4": {  
    "precision": 0.0,  
    "recall": 0.0,  
    "f1": 0.0,  
    "support": 0  
  }  
},  
"averages": {  
  "micro": {  
    "precision": 0.75,  
    "recall": 0.75,  
    "f1": 0.75  
  },  
  "macro": {  
    "precision": 0.40625,  
    "recall": 0.40909090909090906,  
    "f1": 0.38675213675213677  
  },  
  "weighted": {  
    "precision": 0.8828125,  
    "recall": 0.75,  
    "f1": 0.7751068376068376  
  }  
},  
"accuracy": 0.75,  
"support_total": 16  
}  
...
```

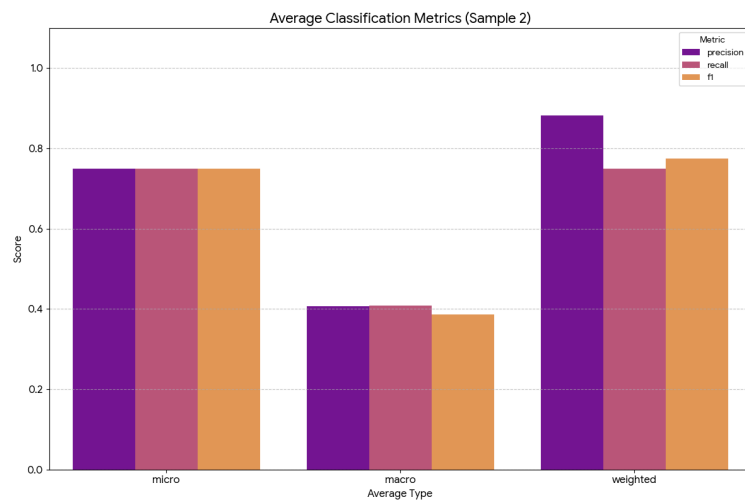
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	5	0	0	0
Q2	3	7	0	1
Q3	0	0	0	0
Q4	0	0	0	0

Precision, Recall and F1





Sample 3

Classification Metrics

...

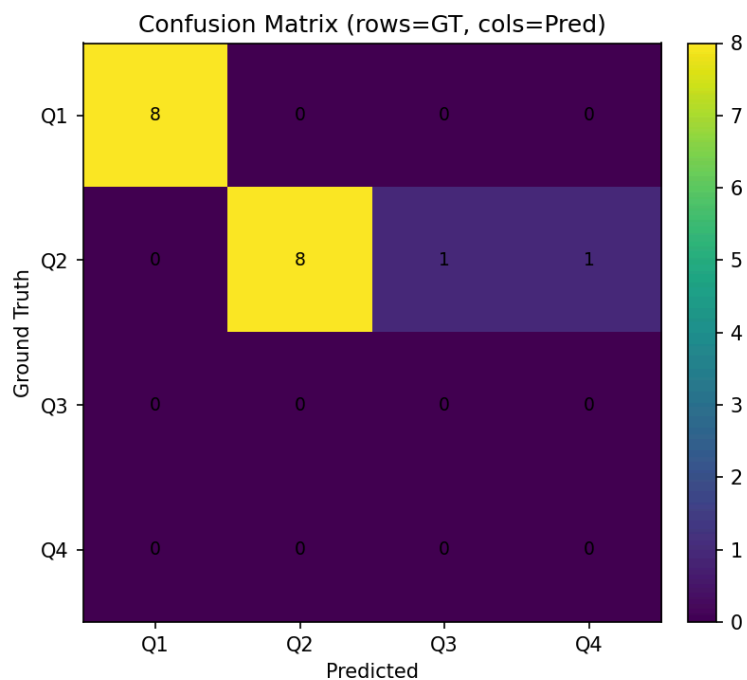
```
{
  "per_class": {
    "Q1": {
      "precision": 1.0,
      "recall": 1.0,
      "f1": 1.0,
      "support": 8
    },
    "Q2": {
      "precision": 1.0,
      "recall": 0.8,
      "f1": 0.8888888888888889,
      "support": 10
    },
    "Q3": {
      "precision": 0.0,
      "recall": 0.0,
      "f1": 0.0,
      "support": 0
    },
    "Q4": {
      "precision": 0.0,
      "recall": 0.0,
      "f1": 0.0,
      "support": 0
    }
  },
  "averages": {
```

```

"micro": {
  "precision": 0.8888888888888888,
  "recall": 0.8888888888888888,
  "f1": 0.8888888888888888
},
"macro": {
  "precision": 0.5,
  "recall": 0.45,
  "f1": 0.4722222222222222
},
"weighted": {
  "precision": 1.0,
  "recall": 0.8888888888888888,
  "f1": 0.9382716049382717
}
},
"accuracy": 0.8888888888888888,
"support_total": 18
}
...

```

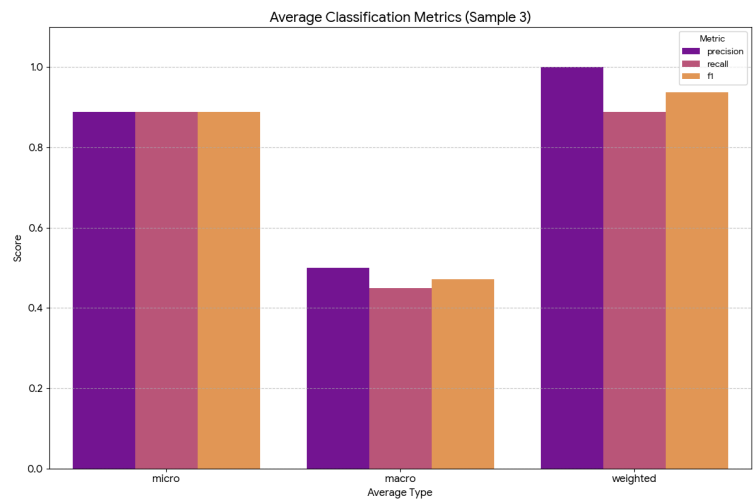
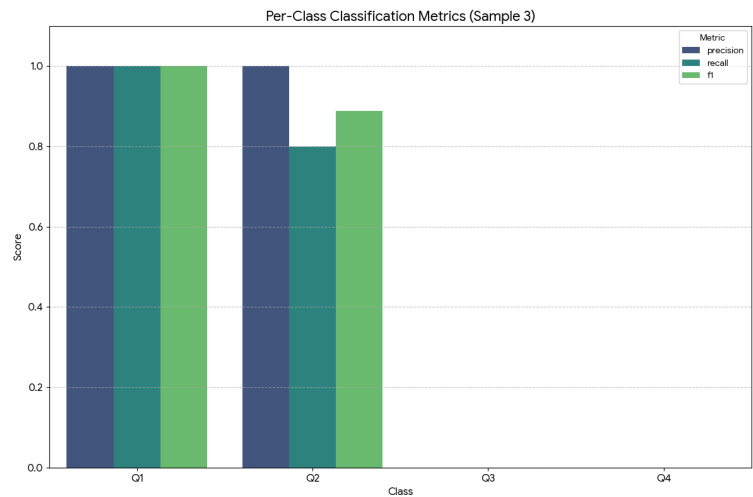
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	8	0	0	0
Q2	0	8	1	1

Q3	0	0	0	0
Q4	0	0	0	0

Precision, Recall and F1



Sample 5

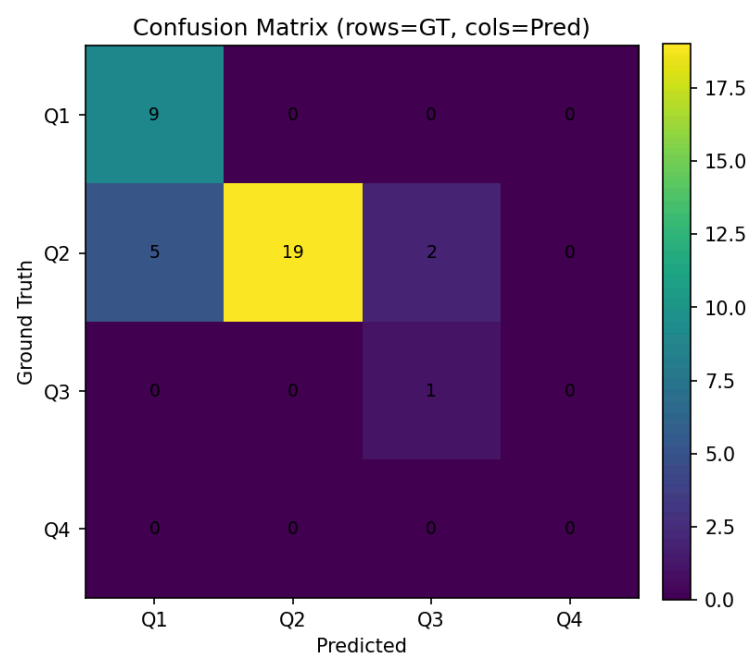
Classification Metrics

...

```
{
  "per_class": {
    "Q1": {
      "precision": 0.6428571428571429,
      "recall": 1.0,
      "f1": 0.782608695652174,
      "support": 9
    }
  }
}
```

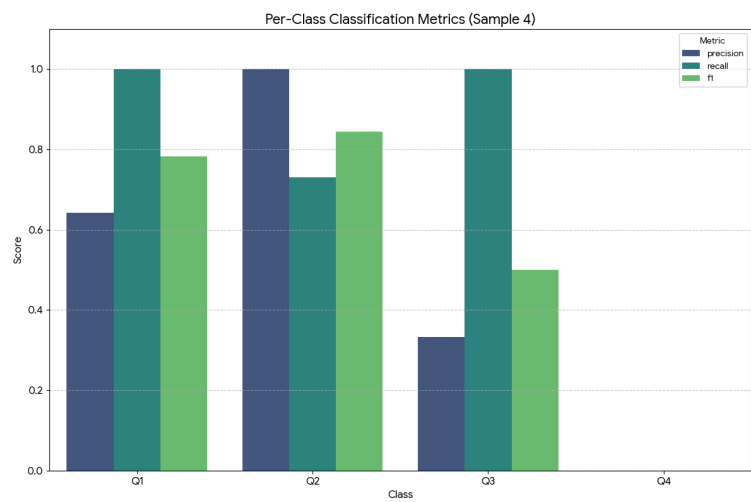
```
},
"Q2": {
  "precision": 1.0,
  "recall": 0.7307692307692307,
  "f1": 0.8444444444444443,
  "support": 26
},
"Q3": {
  "precision": 0.3333333333333333,
  "recall": 1.0,
  "f1": 0.5,
  "support": 1
},
"Q4": {
  "precision": 0.0,
  "recall": 0.0,
  "f1": 0.0,
  "support": 0
}
},
"averages": {
  "micro": {
    "precision": 0.8055555555555556,
    "recall": 0.8055555555555556,
    "f1": 0.8055555555555556
  },
  "macro": {
    "precision": 0.494047619047619,
    "recall": 0.6826923076923077,
    "f1": 0.5317632850241546
  },
  "weighted": {
    "precision": 0.8921957671957672,
    "recall": 0.8055555555555556,
    "f1": 0.8194176060118088
  }
},
"accuracy": 0.8055555555555556,
"support_total": 36
}
...
```

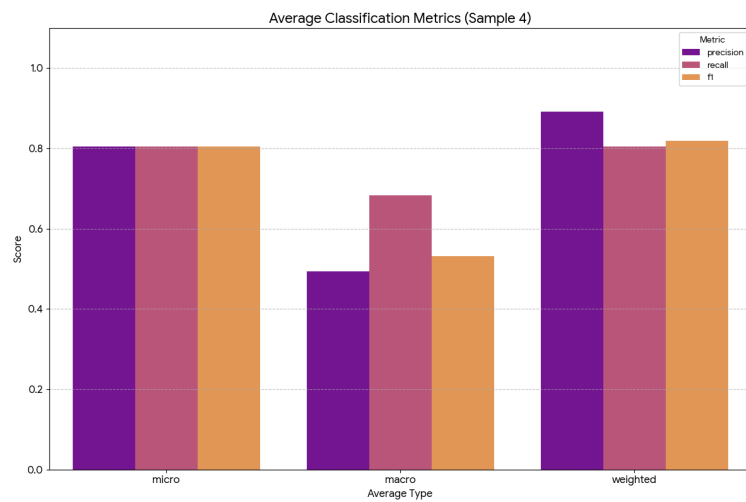
Confusion Matrix



GT\PRED	Q1	Q2	Q3	Q4
Q1	9	0	0	0
Q2	5	19	2	0
Q3	0	0	1	0
Q4	0	0	0	0

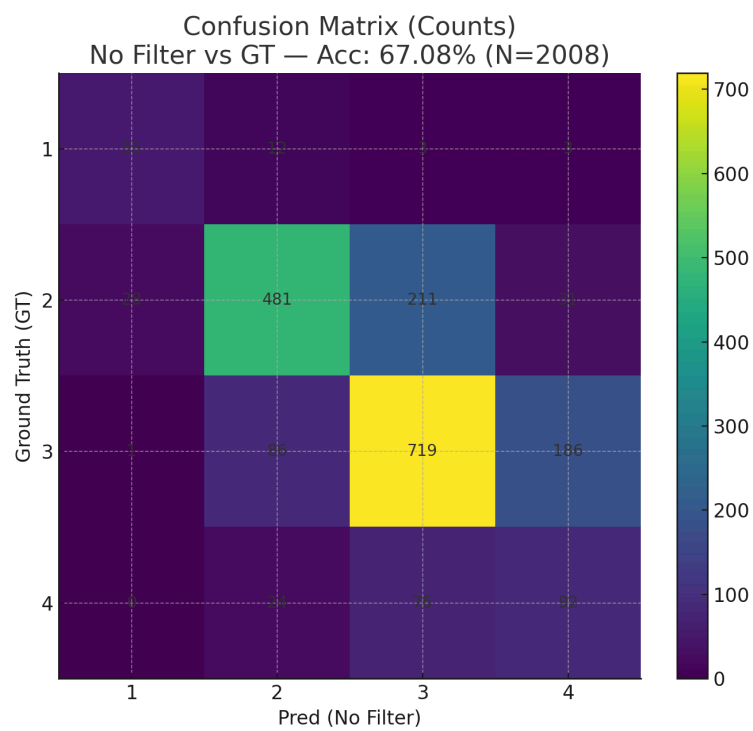
Precision, Recall and F1





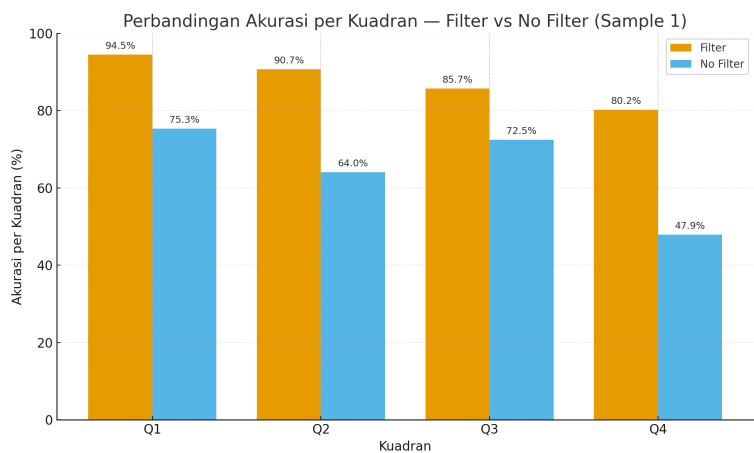
Sample 01 No Filter

Confusion Matrix



GT/Pred	1	2	3	4
1	55	12	3	3
2	28	481	211	31
3	1	86	719	186
4	0	24	76	92

Comparison



Quadra nt	Recall_Filter_ %	Recall_NoFilter_ %	Diff_NoFilter_minus_Filter_ pp	GT_Support_ N
Q1	94.52	75.34	-19.18	73
Q2	90.68	64.05	-26.63	751
Q3	85.69	72.48	-13.21	992
Q4	80.21	47.92	-32.29	192

Uji Ketahanan dan Keandalan

CPU

CPU 1 Camera Pstats

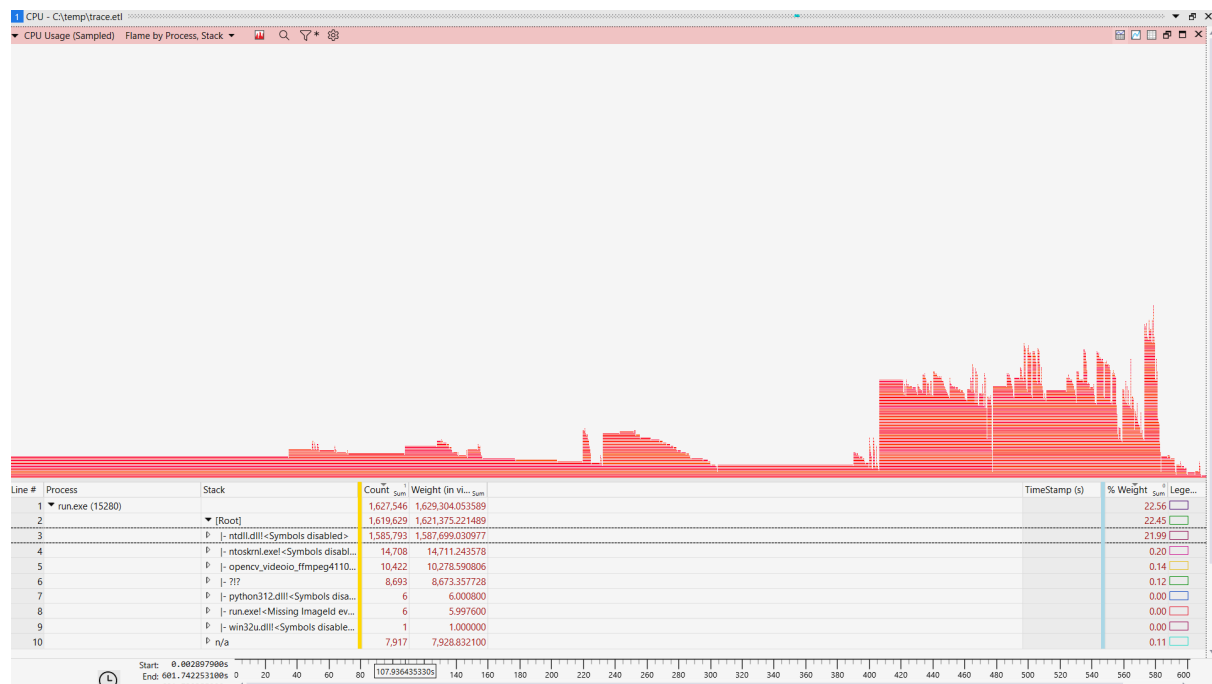
func_label	ncall s	cumtim e (s)	percall (ms)	grou p	description
stdlib/queue.py - get	71	73.5	13.96	stdli b	Ambil item dari Queue secara blocking
stdlib/~ - <built-in method builtins.exec>	191	56	0.01	stdli b	Interpreter mengeksekusi string kode Python.

stdlib/ __init__.py - <module>	1	56	16.71	stdli b	Inisialisasi paket utama aplikasi EyeLog.
stdlib/~ - <built-in method exec>	1	54.4	9577.5 3	stdli b	Fungsi bawaan untuk menjalankan blok kode dinamis.
stdlib/threading.py - wait	81	45.3	3.35	stdli b	Thread menunggu Event/Condition hingga dipicu atau timeout.
app/camera_detail.py - closeEvent	1	35.5	0.08	app	Cleanup lengkap saat jendela detail kamera ditutup, termasuk stop StreamWorker.
app/coverage_logger.py - stop	1	35.5	1.59	app	Menghentikan logger asinkron dan flush CSV log coverage.
stdlib/threading.py - join	1	35.5	0.04	stdli b	Menunggu thread target selesai sebelum melanjutkan.
stdlib/~ - <method 'join' of '_thread._ThreadHandle' objects>	1	35.5	0.24	stdli b	Lower level join handle thread di C API.
stdlib/threading.py - _bootstrap	1	35.5	0.01	stdli b	Rutinitas internal untuk memulai thread Python.
stdlib/threading.py - _bootstrap_inner	1	35.5	0.07	stdli b	Bagian inti bootstrap yang menjalankan target function.
app/stream_worker.py - _iteration	382	29.7	0.93	app	Loop periodik QTimer: baca & proses frame, cek state, handle reconnect.
dep/~ - <method 'acquire' of '_thread.lock' objects>	360	28.1	23.51	dep	Mengunci mutex/kunci thread safe.
dep/~ - <built-in method msleep>	115	9.5	61.09	dep	Tidur dalam milidetik di thread Qt (QThread.msleep).
app/stream_worker.py - _process_frame	250	8.9	1.68	app	Pipeline frame + ekstraksi foreground +

					kontur + metrik coverage.
app/roi_window.py - update_preview	1406	8.6	0.13	app	Refresh tampilan pratinjau ROI di overlay UI.
dep/~ - <method 'exec' of 'PySide6.QtWidgets.QDialog' objects>	4	4.7	1156.69	dep	Menampilkan dialog Qt secara modal & tunggu respon.
app/material_detector.py - process_frame	250	4.1	0.32	app	Lakukan backgroundâ subtraction + morfologi untuk hasilkan mask biner.
app/stream_worker.py - _read_with_timeout	266	3.8	0.03	app	Membaca frame cv2.VideoCapture dengan batas waktu aman.
dep/~ - <built-in method information>	6	3.6	512.71	dep	QMessageBox.informati on kotak dialog info.
app/main_window.py - _on_edit_camera_data_loaded	1	3.5	0.09	app	Callback: buka dialog edit setelah data kamera di fetch
dep/~ - <built-in method fromImage>	1656	3.1	1.89	dep	Konversi QImage QPixmap (Qt).
dep/~ - <method 'copy' of 'numpy.ndarray' objects>	2760	3	0.96	dep	Duplikasi matriks NumPy tanpa pengaruh aslinya.
dep/~ - <cvtColor>	2406	2.5	0.74	dep	OpenCV cvtColor konversi ruang warna.
dep/~ - <method 'apply' of 'cv2.BackgroundSubtractorMO G2' objects>	250	2.4	7.02	dep	OpenCV MOG2 apply() hasilkan mask foreground.
app/material_detector.py - process_mask	250	2.3	0.2	app	Analisis mask biner hitung kontur & statistik coverage.
app/preview_scheduler.py - run	1	2	0.14	app	Thread SnapshotWorker:

					jadwalkan & ambil snapshot kamera.
app/preview_scheduler.py - get_capture	1	2	0.03	app	Ambil VideoCapture dari pool reuse atau buat baru.
app/preview_scheduler.py - _create_capture	1	2	2003.2	app	Buat VideoCapture baru dengan pengaturan spesifik.
app/roi_selector.py - _update_display	141 6	1.9	0.04	app	Render ulang ROI yang dipilih ke layar selama seleksi.

CPU 3 Camera Trace Windows Performance Record



Line #	Stack	% Weight
1	run.exe (15280)	22.56%
2	[Root]	22.45%
3	ntdll.dll!<Symbols disabled>	21.99%
4	ntoskrnl.exe!<Symbols disabl..	0.20%
5	opencv_videoio_ffmpeg4110..	0.14%
6	??	0.12%
7	python312.dll!<Symbols disa..	0.00%

8	run.exe!<Missing ImageId ev..	0.00%
9	win32u.dll!<Symbols disable..	0.00%
10	n/a	0.11%



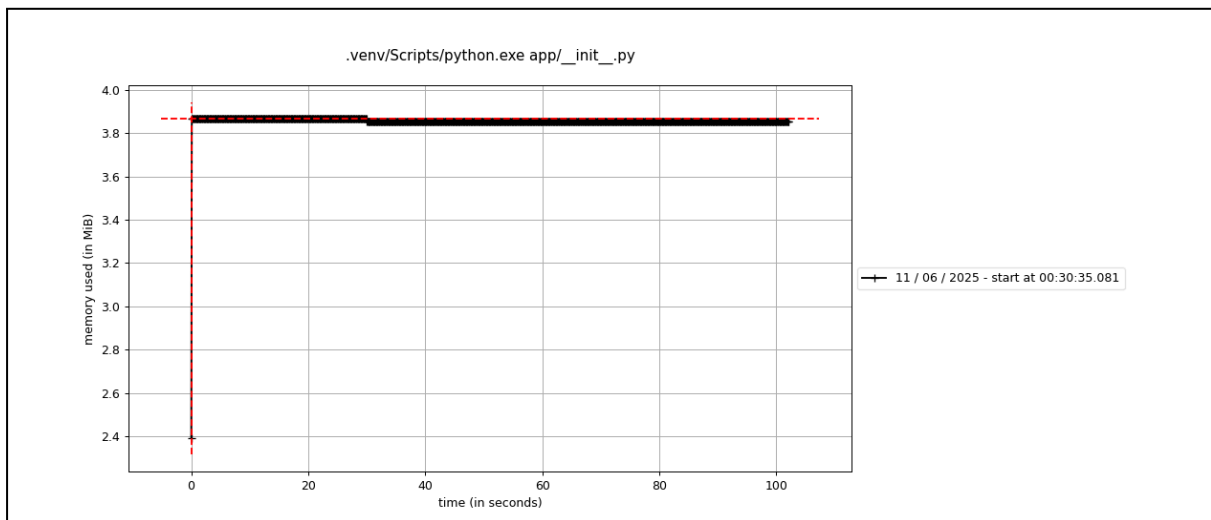
Line #	Process	% Weight
1	run.exe (15280)	22.56%
2	- (Thread ID: 15280)	5.18%
3	- (Thread ID: 3804)	1.58%
4	- (Thread ID: 7508)	1.08%
5	- (Thread ID: 21156)	1.08%
6	- (Thread ID: 20172)	1.08%
7	- (Thread ID: 17432)	1.07%

FPS

source	frames	total_time_s	fps	ms_per_frame
./resource/video_dir/sampl e_01/ROI_01_CUT.mp4	37481	1249.367	30	33.33

Memory

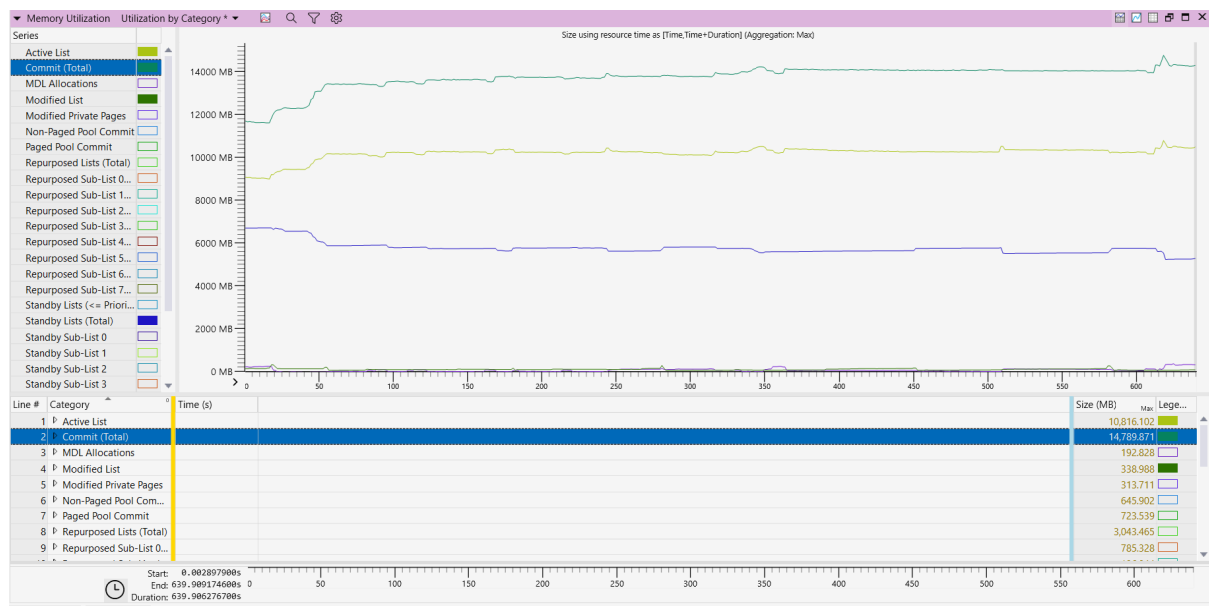
Memory 1 Camera Mprof



Profiling memori menggunakan mprof (interval sampling 0,1 detik) menunjukkan bahwa penggunaan memori aplikasi sangat ringan dan stabil. Selama periode pengujian 100 detik—yang mencakup proses startup, streaming video, pengolahan algoritma background subtraction MOG2, interaksi GUI, hingga proses penutupan—penggunaan RAM tercatat hanya sekitar **3,83 ± 0,02 MB**. Grafik moving-average penggunaan memori memperlihatkan kurva datar, menandakan tidak adanya tren kenaikan memori secara kumulatif yang mengindikasikan memory leak. Penggunaan memori tertinggi selama pengujian tetap di bawah 4 MB, jauh dari ambang batas ideal sekitar 100–200 MB untuk aplikasi desktop ringan. Setelah aplikasi ditutup, penggunaan memori kembali ke baseline interpreter Python (~2,3 MB)—menandakan bahwa semua alokasi memori seperti buffer video telah dilepas dengan sempurna.

Dengan profil memori seoptimal ini, aplikasi dapat dijalankan dengan aman bahkan pada perangkat keras lama seperti laptop dengan Intel Core i5-4300U dan RAM 8 GB, tanpa risiko masalah thrashing atau penggunaan swap berlebihan. Grafik dan data lengkap pengukuran memori dapat dilihat pada [memory_profile.png](#) dan [memory_profile.csv](#).

Memory 3 Camera Windows Performance Recorder



Line #	Category	Size (MB)
1	Active List	10,816.102
2	Commit (Total)	14,789.871
3	MDL Allocations	192.828
4	Modified List	338.988
5	Modified Private Pages	313.711
6	Non-Paged Pool Commit	645.902
7	Paged Pool Commit	723.539
8	Repurposed Lists (Total)	3,043.465
9	Repurposed Sub-List 0 (Delta)	785.328
10	Repurposed Sub-List 1 (Delta)	196.844
11	Repurposed Sub-List 2 (Delta)	354.766
12	Repurposed Sub-List 3 (Delta)	0.000
13	Repurposed Sub-List 4 (Delta)	145.926
14	Repurposed Sub-List 5 (Delta)	1,560.602

15	Repurposed Sub-List 6 (Delta)	0.000
16	Repurposed Sub-List 7 (Delta)	0.000
17	Standby Lists (<= Priority 5)	6,653.402
18	Standby Lists (Total)	6,731.398
19	Standby Sub-List 0	195.395
20	Standby Sub-List 1	34.500
21	Standby Sub-List 2	95.453
22	Standby Sub-List 3	0.000
23	Standby Sub-List 4	42.055
24	Standby Sub-List 5	6,617.613
25	Standby Sub-List 6	8.609
26	Standby Sub-List 7	78.223
27	Total Physical Memory	16,206.00 0
28	Zero and Free Lists	398.063

Unit Test

```
PC-Windows@DESKTOP-U88U1AD MINGW64 /c/Users/PC-Windows/Documents
/Github/opencv-py-pdu/app (master)
$ pytest
===== test session starts =====
platform win32 -- Python 3.13.2, pytest-8.4.0, pluggy-1.6.0
rootdir: C:\Users\PC-Windows\Documents\Github\opencv-py-pdu\app
collected 13 items

tests\test_camera_roi.py ..                [ 15%]
tests\test_camera_url.py ...               [ 38%]
tests\test_database.py .....              [ 76%]
tests\test_material_detector.py ...        [100%]

===== 13 passed in 0.58s =====
(.venv)
PC-Windows@DESKTOP-U88U1AD MINGW64 /c/Users/PC-Windows/Documents
/Github/opencv-py-pdu/app (master)
$ █
```

No	Nama Test
1	test_camera_roi_saves_correct_roi
2	test_camera_roi_remove
3	test_camera_url_validation_valid
4	test_camera_url_parse_host_and_port
5	test_insert_and_fetch_camera
6	test_foreground_detects_object
7	test_roi_handling_per_camera
8	test_contour_detection_on_foreground
9	test_foreground_blank_returns_zero
10	test_update_camera_data